Freddy Zhou

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EDUCATION

Stony Brook University **Aug. 2024 - Present**
Bachelor of Science in Computer Science | Expected Graduation: May 2028 *Stony Brook, NY*
GPA: 3.94/4.00
Related Coursework: Software Development, Systems Fundamentals I, System Fundamentals II, Analysis of Algorithms, Data Structures & Algorithms, Foundations of Computer Science, & Programming Abstractions

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, C++, C, HTML, CSS, OCaml, SQL, Swift
Frameworks & Libraries: React.js, Node.js, Spring Boot, Express.js, Material UI
Technologies & Tools: Git, GitHub, GitHub Actions, MySQL, PostgreSQL, MongoDB, Docker, Gradle
Certifications: AWS Certified Cloud Practitioner - In Progress

EXPERIENCE

StayHome Healthcare (Startup) **May 2026 - Present**
Software Engineer *New York, NY*

- Owned end-to-end development of a full stack telehealth system by designing and implementing React client-side features and Spring Boot backend services through building REST APIs, optimizing request caching, and validating API response data.
- Eliminated manual build verification by establishing a continuous integration (CI) pipeline using GitHub Actions and Docker to create an automated workflow to run the Gradle builds and 20+ integration tests on every pull request.
- Eliminated server-side session storage by implementing a stateless JWT-based authentication service through employing Spring Security, building custom authentication filters, and designing business logic for authentication services like token validation.

Handshake AI **Jun. 2026 - Present**
Software Engineering AI Trainer *New York, NY (Remote)*

- Evaluated and scored multi-turn code completions from frontier large language models (LLMs) spanning across diverse types of software engineering domains, including frontend and backend development, API integration, algorithm design, and debugging.
- Increased the number of tasks added to the production data pipeline through achieving a quality score increase of 10% and approval rate increase of 30% by implementing a systematic approach for producing intricate technical rationales.
- Reviewed model-generated completions built with modern web development technologies, including React.js, Next.js, and TypeScript, by assessing technical clarity, technical communication, and response quality.

Google x Basta **Mar. 2026 - Present**
Fellow *New York, NY*

- Solved 15+ new advanced data structures and algorithms problems by closely collaborating with a Google Software Engineer mentor on 1:1 weekly calls going deep into a variety of technical concepts.
- Reinforced pattern recognition and problem solving skills by collaborating in realistic weekly technical mock interviews with a Google Software Engineer who provided feedback on algorithmic problem solving and time and space complexity analysis.

Stony Brook University **Jan. 2026 - May 2026**
Object-Oriented Programming Teaching Assistant *Stony Brook, NY*

- Facilitated two weekly 90-minute laboratory sessions for 30+ students where students' course required skills were refined by presenting core material through problem sets and by providing hands-on assistance.
- Mentored 20+ students during weekly 4-hour office hours by optimizing students' foundational proficiency in object-oriented principles and providing guidance in solving complex programming problems.

PROJECTS

Playlister (Spotify Clone) | *JavaScript, React.js, Node.js, Express.js, PostgreSQL* **Dec. 2025 - Mar. 2026**

- Built a full stack web application utilizing a React frontend and an Express backend, allowing users to save 1,000+ songs while maintaining performance by employing optimized queries to manipulate data using Sequelize.
- Minimized potential user induced errors by implementing foolproof UI logic that provides safeguards through disabling certain operations for numerous scenarios.
- Eliminated prop drilling from 3+ component layers by leveraging the use of React context and hooks.

Library REST API | *Java, Spring Boot, PostgreSQL* **Jul. 2025 - Aug. 2025**

- Developed an efficient RESTful API that supports all CRUD operations, optimizing customer experiences by achieving over a 99% endpoint reliability rate by performing rigorous integration testing.
- Reduced user wait time for operations related to retrieving data from the server by 88% (from 44ms to 5ms) by employing the pagination design pattern to endpoints that handle incoming GET requests.
- Allowed for future transition to microservices by implementing a layering architecture, rather than a monolithic architecture.

HONORS & AWARDS

Dean's List **2024-Present**